

Olympiad Test Paper — Science (Class 7)

Chapter 7: Weather, Climate and Adaptations of Animals to Climate

Subject	Science (NCERT Class 7)
Chapter	7 — Weather, Climate and Adaptations of Animals to Climate
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Remember the key idea throughout: weather is the day-to-day state of the atmosphere, while climate is its average over many years. Some questions give a small weather-report table — read the numbers carefully before answering. Most questions need two to four reasoning steps; watch for traps such as confusing weather with climate. No calculator is needed. Do not write on this paper — mark answers on your answer sheet.

1. A TV presenter says, "Tomorrow Delhi will be 32 °C with light rain in the evening." This statement describes:

- (A) the altitude of Delhi
- (B) the humidity of Delhi only
- (C) the weather of Delhi
- (D) the climate of Delhi

2. Which statement correctly distinguishes weather from climate?

- (A) weather is the long-term average; climate changes hour to hour
- (B) weather is the day-to-day state; climate is the average over many years
- (C) weather and climate mean exactly the same thing
- (D) climate is measured with a thermometer; weather is not measured at all

3. The average weather of a place is usually taken over a period of at least:

- (A) twenty-five years
- (B) one month
- (C) one year
- (D) one week

4. On one winter day it snowed heavily in a hill town. A visitor said, "This proves the town's climate has become colder." The best reply is:

- (A) correct, because snow always means a colder climate
- (B) correct, because climate changes from day to day

- (C) wrong, because snow has nothing to do with temperature
- (D) wrong, because a single day's weather does not establish climate

5. Mumbai receives heavy rain every June to September, year after year. This repeated yearly pattern is part of Mumbai's:

- (A) weather for one day
- (B) climate
- (C) altitude
- (D) wind speed only

6. The scientific study of weather is called:

- (A) meteorology
- (B) astronomy
- (C) geology
- (D) biology

7. Which of the following can change within a few hours?

- (A) the climate of a desert
- (B) the altitude of a hill station
- (C) the latitude of a place
- (D) the weather of a city

8. Chennai stays warm and humid almost all year, while Delhi has hot summers and cold winters. This long-term contrast is best described as a difference in their:

- (A) climate
- (B) weather on one particular day
- (C) cloud colour
- (D) wind direction for one hour

9. A farmer in Punjab chooses to sow wheat in winter and rice in the monsoon. This decision is based mainly on the region's:

- (A) weather on the day of sowing
- (B) climate
- (C) soil colour only
- (D) wind speed at noon

10. Which pair lists two factors that help decide the climate of a place?

- (A) the colour of houses and the number of cars
- (B) altitude and distance from the sea
- (C) the day of the week and the time of day
- (D) a single afternoon's cloud cover

11. Shimla is much cooler than the plains nearby, even in summer. The main reason is that Shimla has a high:

- (A) humidity
- (B) wind speed
- (C) altitude
- (D) rainfall on one day

12. Coastal cities such as Mumbai have milder temperatures than inland cities at the same latitude. This is because the nearby sea:

- (A) increases the altitude
- (B) blocks all sunlight
- (C) moderates the temperature, reducing extremes
- (D) removes all humidity from the air

13. As you go higher up a mountain, the temperature generally:

- (A) increases
- (B) stays exactly the same
- (C) first rises and then becomes zero
- (D) decreases

14. The Indian monsoon brings heavy rain because seasonal winds carry large amounts of moisture from the:

- (A) Indian Ocean
- (B) Himalayan snow
- (C) Thar Desert
- (D) dry plateau

15. Two towns lie at the same latitude. Town P is on the coast; Town Q is far inland. Compared with Q, town P is most likely to have:

- (A) much greater day-to-night temperature swings
- (B) no rainfall at all
- (C) smaller temperature swings and higher humidity
- (D) a higher altitude

16. The ultimate source of energy that drives almost all of Earth's weather is:

- (A) the Moon
- (B) the Sun
- (C) the wind
- (D) ocean currents alone

17. Weather is often hard to predict far in advance mainly because:

- (A) thermometers cannot measure temperature
- (B) the Sun stops shining at night
- (C) climate never changes
- (D) the atmosphere is a constantly changing system with many interacting factors

18. During a single day the air temperature is usually lowest:

- (A) just before sunrise
- (B) in the early afternoon
- (C) at noon
- (D) at midnight exactly, always

19. Reliable day-to-day weather forecasts are typically trustworthy only for about:

- (A) the next hundred years
- (B) the next ten years
- (C) the next few days
- (D) no period at all

20. On a clear day the air over land usually becomes warmest in the early afternoon rather than exactly at noon because:

- (A) the Sun is closest to Earth then
- (B) the Moon adds heat in the afternoon
- (C) humidity always falls to zero at noon
- (D) the ground keeps absorbing heat for a while after the Sun is highest

21. Which instrument is used to measure the highest and lowest air temperatures reached during a day?

- (A) a rain gauge
- (B) an anemometer
- (C) a barometer only
- (D) a maximum–minimum thermometer

22. The amount of rainfall at a place is measured using a:

- (A) maximum–minimum thermometer
- (B) rain gauge
- (C) weighing balance
- (D) measuring tape on the ground

23. A rain gauge collects 25 mm of rain on Monday and 15 mm on Tuesday. The total rainfall over the two days is:

- (A) 10 mm
- (B) 20 mm
- (C) 375 mm
- (D) 40 mm

24. The "maximum temperature" reported for a day means the:

- (A) average of all readings that day
- (B) highest temperature reached during that day
- (C) temperature at exactly midnight
- (D) lowest temperature reached during that day

25. Humidity in a weather report is usually expressed as a:

- (A) length in millimetres
- (B) temperature in degrees Celsius
- (C) percentage
- (D) speed in km/h

26. On a weather chart, the difference between the day's maximum and minimum temperature is called the:

- (A) temperature range
- (B) rainfall
- (C) humidity
- (D) wind speed

27. A maximum–minimum thermometer on a certain day reads: maximum 34 °C, minimum 22 °C. The temperature range for that day is:

- (A) 56 °C
- (B) 22 °C
- (C) 12 °C
- (D) 28 °C

28. Which weather element does a rain gauge NOT measure?

- (A) the depth of collected rainwater
- (B) how much rain fell
- (C) rainfall in millimetres
- (D) the air temperature

29. Consider this weather record for one week:

Day : Mon Tue Wed Thu Fri

Max (°C) : 30 31 33 36 34

Min (°C) : 22 23 24 22 25

On which day was the temperature range the largest?

- (A) Monday
- (B) Wednesday
- (C) all days had equal range
- (D) Thursday

30. Using the same table as the previous question, the highest maximum temperature of the week was recorded on:

- (A) Monday
- (B) Thursday
- (C) Tuesday
- (D) Friday

31. A week's record shows daily rainfall (mm): Mon 0, Tue 12, Wed 0, Thu 30, Fri 8. The total rainfall for the week was:

- (A) 30 mm
- (B) 42 mm
- (C) 10 mm
- (D) 50 mm

32. From the rainfall record in the previous question (Mon 0, Tue 12, Wed 0, Thu 30, Fri 8), the day with the heaviest rainfall was:

- (A) Tuesday
- (B) Friday
- (C) Monday
- (D) Thursday

33. A humidity table reads: 06:00 → 90%, 12:00 → 55%, 18:00 → 70%. The humidity was lowest:

- (A) at 06:00
- (B) at noon
- (C) at 18:00
- (D) it stayed constant

34. A weather report gives: max 28 °C, min 18 °C, humidity 85%, rainfall 12 mm. Which figure tells you how hot it got?

- (A) the humidity, 85%
- (B) the maximum temperature, 28 °C
- (C) the rainfall, 12 mm
- (D) the minimum temperature, 18 °C

35. Over four days a city's maximum temperatures were 30, 32, 34 and 36 °C. The average maximum temperature was:

- (A) 33 °C
- (B) 30 °C
- (C) 36 °C
- (D) 132 °C

36. A place that is hot all year, with very heavy rainfall and dense forests, has a climate best described as:

- (A) hot and dry (desert)
- (B) polar (very cold)
- (C) hot and wet (tropical)
- (D) cold and dry

37. The polar regions are characterised by:

- (A) heavy rain and dense rainforest
- (B) extreme cold and ice for most of the year
- (C) hot dry sand dunes
- (D) mild temperatures all year

38. A desert region with very high daytime temperatures and almost no rainfall has a climate best called:

- (A) polar
- (B) hot and dry
- (C) tropical rainforest
- (D) cold and humid

39. Of the following, the region with the greatest variety (biodiversity) of plants and animals is usually the:

- (A) tropical rainforest
- (B) polar ice cap
- (C) hot sandy desert
- (D) high mountain snowfield

40. Which set of conditions is typical of a tropical rainforest climate?

- (A) low temperature and heavy snow (B) high temperature and almost no rain
(C) high temperature and heavy rainfall (D) low temperature and very dry air

41. A polar bear has a thick layer of fat (blubber) under its skin mainly to:

- (A) help it float and feel lighter (B) make it run faster on land
(C) insulate its body and store energy in the cold
(D) attract a mate by its colour

42. The white fur of a polar bear is most useful because it:

- (A) keeps the bear cool in summer heat (B) helps it blend with the snow while hunting
(C) makes the bear easy for prey to see (D) reflects away all the bear's body heat

43. Polar bears have relatively small ears and short tails. The main advantage of these small extremities is that they:

- (A) reduce heat loss from the body (B) improve the bear's hearing in storms
(C) help the bear swim in a straight line (D) increase the surface area exposed to cold

44. Penguins in very cold regions often huddle together in large groups. This behaviour mainly helps them to:

- (A) catch fish more easily on land (B) fly long distances
(C) cool down in the heat (D) reduce heat loss and keep warm

45. A polar bear has wide, large paws. These are most useful for:

- (A) digging deep underground burrows (B) climbing tall trees
(C) spreading its weight on snow and gripping ice
(D) flying short distances

46. Which combination of features would best suit an animal living in a polar climate?

- (A) thin fur, large ears, dark colour (B) no fur, very long tail, bright colour
(C) thick fur, small ears, white colour, layer of fat
(D) thin skin, wide green stripes, webbed wings

47. Many rainforest monkeys have long, strong tails and limbs. The main reason is to:

- (A) climb and swing among trees (B) keep warm in freezing cold
(C) store fat for winter (D) camouflage against white snow

48. A red-eyed tree frog and many rainforest insects are bright green. This colouring mostly helps them to:

- (A) blend in with the green leaves (B) stay warm in the cold
(C) glow in total darkness (D) float on water

49. The toucan, a rainforest bird, has a very large beak that helps it:

- (A) survive freezing polar winters (B) reach and feed on fruit on thin branches
(C) dig burrows in dry sand (D) swim underwater for fish

50. The lion-tailed macaque is well known as an animal that lives in the:

- (A) polar ice caps
- (B) hot sandy deserts
- (C) tropical rainforests of the Western Ghats
- (D) cold high mountain snowfields

51. In dense rainforests where it is hard to see far, many animals such as howler monkeys and parrots communicate using:

- (A) bright body heat
- (B) white winter fur
- (C) thick layers of fat
- (D) loud calls and sounds

52. Migration is best described as the:

- (A) growth of fur in winter on one animal
- (B) change of an animal's colour during the day
- (C) seasonal long-distance travel of animals to escape harsh climate or find food
- (D) building of a permanent nest in one place forever

53. Siberian cranes are seen at Bharatpur in India during winter. The best explanation is that the cranes:

- (A) migrate from colder Siberia to escape its harsh winter
- (B) were born in India and never leave
- (C) change into a different species in winter
- (D) come to India only when it is summer there

54. The main trigger that tells migratory birds it is time to travel is usually a change in:

- (A) temperature, day length, and food availability
- (B) the colour of the leaves on one tree
- (C) the price of grain in markets
- (D) the number of people watching them

55. Migration is a useful strategy mainly because, instead of enduring a harsh season in one place, the animal:

- (A) stops eating until the season passes
- (B) moves to a region where conditions are more favourable
- (C) grows much larger overnight
- (D) turns invisible to predators

56. An animal has a thick coat of fur, a small surface area compared with its body size, and white colouring. It is best adapted to live in a:

- (A) polar region
- (B) tropical rainforest
- (C) hot dry desert
- (D) shallow warm lake

57. A monkey with grasping hands and a long tail is suddenly placed in a polar ice region. It would most likely struggle because its body is adapted for:

- (A) freezing cold and snow
- (B) flying over oceans
- (C) burrowing in hot sand
- (D) climbing trees in a warm, wet forest

58. Why do polar animals tend to have a small surface-area-to-volume ratio (a rounder, compact body) rather than a long, thin one?

- (A) a compact body loses heat more slowly, helping survival in the cold
- (B) a compact body loses heat faster, which is helpful in the cold
- (C) it makes them taller than predators
- (D) it has no effect on heat at all

59. Which of the following is the clearest example of an adaptation to a HOT, WET tropical climate rather than a cold one?

- (A) a thick layer of blubber under the skin
- (B) white fur for snow camouflage
- (C) strong limbs and a grasping tail for climbing wet trees
- (D) small ears to reduce heat loss

60. Four animals are described by a single feature each. Which feature would be LEAST helpful for surviving in a polar climate?

- (A) a thick double layer of fur
- (B) a wide paw that spreads weight on snow
- (C) a layer of fat under the skin
- (D) very large, thin ears with a big surface area

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 7: Weather, Climate and Adaptations of Animals to Climate

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	C	21	D	31	D	41	C	51	D
2	B	12	C	22	B	32	D	42	B	52	C
3	A	13	D	23	D	33	B	43	A	53	A
4	D	14	A	24	B	34	B	44	D	54	A
5	B	15	C	25	C	35	A	45	C	55	B
6	A	16	B	26	A	36	C	46	C	56	A
7	D	17	D	27	C	37	B	47	A	57	D
8	A	18	A	28	D	38	B	48	A	58	A
9	B	19	C	29	D	39	A	49	B	59	C
10	B	20	D	30	B	40	C	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) "Tomorrow ... 32 °C with light rain in the evening" describes the day-to-day state of the atmosphere — that is weather. Climate would be the long-term average; altitude and humidity-only are red herrings.

2. (B) Weather is the short-term, day-to-day state of the atmosphere; climate is its average taken over many years. Option (A) reverses the two; (C) wrongly equates them; (D) is false — both are measured.

3. (A) By convention, climate is the average weather taken over a long period, usually at least 25 years (often quoted as ~30). A week, month or single year is far too short to smooth out year-to-year weather variation.

4. (D) A single day's snowfall is one weather event; climate is the long-term average, so one cold day cannot establish that the climate has changed. (A) and (B) wrongly treat one day as climate; (C) confuses weather's daily change with climate.

- 5. (B)** A pattern that repeats year after year (heavy June–September rain) is part of the long-term average — Mumbai's climate. One day's rain would be weather; altitude and wind speed are single factors, not the overall pattern.
- 6. (A)** The scientific study of weather is meteorology (its practitioners are meteorologists). Geology is rocks/Earth, astronomy is space, biology is living things.
- 7. (D)** Weather can change within hours. Climate (A) changes only over decades; altitude (B) and latitude (C) of a place are essentially fixed.
- 8. (A)** Comparing Chennai (warm, humid all year) with Delhi (hot summers, cold winters) over the long run is a comparison of climate. The other options describe single-moment weather details, not the long-term pattern.
- 9. (B)** Crop choices for a whole season rest on the region's typical long-term conditions — its climate — not on the weather of the single day of sowing, nor soil colour or noon wind speed.
- 10. (B)** Altitude and distance from the sea are genuine climate-shaping factors (higher = cooler; coastal = milder). House colour, car count, weekday and one afternoon's cloud cover do not determine a region's climate.
- 11. (C)** Temperature falls with height (about 6.5 °C per km), so a high-altitude town like Shimla is cooler than the nearby plains. Humidity, wind speed and one day's rainfall do not explain a steady year-round coolness.
- 12. (C)** The sea heats and cools slowly, so it moderates the air temperature of nearby land, reducing the gap between hot and cold extremes. It does not raise altitude, block all sunlight, or remove all humidity (in fact coasts are more humid).
- 13. (D)** As altitude increases, temperature generally decreases. So the answer is "decreases"; it does not rise, stay constant, or behave like (C).
- 14. (A)** The Indian monsoon winds pick up huge amounts of moisture from the warm Indian Ocean and bring rain over land. The Thar Desert and dry plateau are dry sources; Himalayan snow does not feed the monsoon rains.
- 15. (C)** Because the sea moderates temperature and adds moisture, the coastal town P has smaller day-to-night temperature swings and higher humidity than inland Q. (A) reverses this; coasts still get rain (B is false); being on the coast does not mean higher altitude (D).
- 16. (B)** The Sun is the ultimate source of the energy that heats air, land and sea, drives winds and the water cycle — i.e. almost all weather. The Moon, wind and ocean currents are not the original energy source (wind and currents are themselves Sun-driven).

17. (D) Weather is hard to forecast far ahead because the atmosphere is a constantly changing system with many interacting factors, so small differences grow quickly. (A), (B) and (C) are simply false statements.

18. (A) Through a normal day the air is usually coolest just before sunrise, after the ground has lost heat all night, and warmest in the early afternoon. Noon and "exactly midnight, always" are traps; early afternoon is the hottest, not the coolest.

19. (C) Day-to-day weather forecasts are reliable only for roughly the next few days; they cannot be trusted for ten or a hundred years (that is the realm of climate projections, which work on averages, not daily detail).

20. (D) Even though the Sun is highest at noon, the ground keeps absorbing more heat than it loses for a while afterwards, so the air keeps warming and peaks in the early afternoon (thermal lag). The Sun's distance, the Moon, and "humidity falling to zero" are false explanations.

21. (D) A maximum–minimum thermometer records both the highest and the lowest temperatures reached during a day. A rain gauge measures rain, an anemometer measures wind speed, a barometer measures air pressure.

22. (B) Rainfall is measured with a rain gauge, which collects rain in a graduated container and reports it in millimetres. The other instruments measure temperature, mass, or length — not rainfall.

23. (D) Total rainfall = $25 + 15 = 40$ mm. (10 mm subtracts instead of adds; 20 mm averages; $375 = 25 \times 15$ multiplies — all traps.)

24. (B) The day's "maximum temperature" is simply the highest temperature reached during that day — not an average, not the midnight value, and certainly not the minimum.

25. (C) Humidity (relative humidity) is the amount of moisture in the air as a percentage of the maximum the air can hold, so it is reported as a percentage. Millimetres are for rain, °C for temperature, km/h for wind.

26. (A) The gap between the day's highest and lowest temperatures is the temperature range (max – min). Humidity, rainfall and wind speed are different quantities.

27. (C) Temperature range = maximum – minimum = $34 - 22 = 12$ °C. (56 adds the two; 22 and 28 just copy single readings — traps.)

28. (D) A rain gauge measures only rainfall (depth of collected water, in mm). It does NOT measure air temperature — that needs a thermometer. The other three options all describe what a rain gauge does measure.

29. (D) Compute each day's range (max – min): Mon $30 - 22 = 8$, Tue $31 - 23 = 8$, Wed $33 - 24 = 9$, Thu $36 - 22 = 14$, Fri $34 - 25 = 9$. Thursday's range (14) is the largest. (Picking the highest max alone, or assuming all equal, are traps.)

30. (B) The highest maximum temperature in the row 30, 31, 33, 36, 34 is 36°C , recorded on Thursday. (Don't confuse "highest maximum" with "largest range" from the previous question — both happen to fall on Thursday here, but for different reasons.)

31. (D) Total weekly rainfall = $0 + 12 + 0 + 30 + 8 = 50$ mm. (30 takes only the wettest day; 42 forgets the 8; 10 is unrelated — traps.)

32. (D) The heaviest single day is the largest value, 30 mm on Thursday. Tuesday (12) and Friday (8) are smaller; Monday is 0.

33. (B) Reading the table, humidity is 90% at 06:00, 55% at noon, 70% at 18:00. The lowest value, 55%, is at noon — air is often driest in the warm middle of the day. It did not stay constant.

34. (B) "How hot it got" is the highest temperature reached, i.e. the maximum temperature, 28°C . Humidity is moisture, rainfall is rain depth, and the minimum (18°C) tells how cool it got, not how hot.

35. (A) Average = $(30 + 32 + 34 + 36) \div 4 = 132 \div 4 = 33^{\circ}\text{C}$. (36 is just the highest, not the average; 30 the lowest; 132 forgets to divide — traps.)

36. (C) Hot all year + very heavy rainfall + dense forest is the hot-and-wet tropical (rainforest) climate. A desert is hot-and-dry, polar is very cold, and cold-and-dry does not fit "hot" and "heavy rain".

37. (B) Polar regions are marked by extreme cold and ice cover for most of the year. Heavy rain/rainforest, hot sand dunes and "mild all year" describe other climates.

38. (B) Very high daytime temperature with almost no rainfall is the hot-and-dry (desert) climate. Polar is cold, tropical rainforest is wet, and cold-and-humid does not fit "very hot, no rain".

39. (A) Tropical rainforests, with warmth and abundant water all year, support the greatest variety (biodiversity) of life. Polar ice, hot sandy deserts and high snowfields are comparatively species-poor.

40. (C) A tropical rainforest climate combines high temperature with heavy rainfall. The cold/snow options and the "high temperature but almost no rain" (that is desert) options are wrong.

41. (C) Blubber is a thick fat layer that insulates the polar bear against extreme cold and stores energy for lean times. It is not mainly for floating, running speed, or attracting a

mate.

42. (B) White fur lets the polar bear blend with the snow and ice so it can approach prey unseen (camouflage). It does not keep it cool, make it easy to see, or reflect away its own body heat.

43. (A) Small ears and a short tail mean less exposed surface area, so the bear loses less body heat in the cold. (D) is the opposite (and wrong); hearing in storms and "swimming straight" are not the reason.

44. (D) Huddling together puts the birds' warm bodies close, cutting the exposed surface area of each and reducing heat loss, so the group stays warmer. It is not for catching fish, flying, or cooling down (it is already freezing).

45. (C) Wide, large paws spread the bear's weight so it does not sink into soft snow and help it grip slippery ice. They are not for burrowing, tree-climbing, or flying.

46. (C) A polar animal is best served by thick fur, small ears, white colour and a fat layer — all reduce heat loss and aid camouflage. The other combinations (large ears, dark/bright colour, thin skin, green stripes) would lose heat or fail to camouflage in snow.

47. (A) Long strong limbs and a grasping tail let rainforest monkeys climb and swing among the trees where they live and feed. Warmth, fat storage and snow camouflage are cold-climate needs, irrelevant in a hot wet forest.

48. (A) Bright green colouring lets a tree frog or insect blend in with the green leaves (camouflage from predators). It is not for keeping warm, glowing in the dark, or floating.

49. (B) The toucan's large light beak lets it reach fruit on thin branches that could not hold its weight, and handle the fruit. It is not for polar winters, sand-burrowing, or underwater fishing.

50. (C) The lion-tailed macaque is a monkey of the tropical rainforests of the Western Ghats in India. It does not live in polar caps, deserts, or snowfields.

51. (D) In dense forest, vision is blocked by foliage, so howler monkeys, parrots and others rely on loud calls and sounds to communicate over distance. "Body heat", "winter fur" and "fat layers" are not communication methods.

52. (C) Migration is the seasonal long-distance travel of animals to escape harsh climate or to find food and breeding grounds. Growing winter fur, a daily colour change, or staying put in one nest are not migration.

53. (A) Siberian cranes appear in India in winter because they migrate from far colder Siberia to escape its harsh winter, returning when it warms. They are not Indian residents, do not change species, and arrive in India's winter (not summer).

54. (A) Migration is triggered chiefly by changes in temperature, day length, and food availability — the natural cues of an approaching harsh season. Leaf colour on one tree, grain prices and onlookers are not the triggers.

55. (B) Rather than endure a harsh season where it is, the migrating animal moves to a region with more favourable conditions (food, warmth, breeding sites). It does not stop eating, grow overnight, or turn invisible.

56. (A) Thick fur, a small surface area for the body's volume, and white colouring all reduce heat loss and give snow camouflage — classic polar-region adaptations. They would be useless or harmful in a desert, rainforest or warm lake.

57. (D) A monkey's grasping hands and long tail are adaptations for climbing trees in a warm, wet forest, so it is ill-suited to bare, freezing polar ice (no trees, deadly cold, no fur/fat for it). It is not adapted to cold, ocean-flight, or hot-sand burrowing.

58. (A) A rounder, compact body has a smaller surface-area-to-volume ratio, so it loses heat more slowly — an advantage in the cold. (B) is the reversed (false) version; the body shape does affect heat, and it is unrelated to height.

59. (C) Strong limbs and a grasping tail for climbing wet trees is a hot-wet tropical-forest adaptation. Blubber, white snow-fur, and small heat-saving ears are all cold-climate adaptations, the opposite of what the question asks.

60. (D) Very large, thin ears have a big surface area, which would lose heat fast — the worst feature for surviving polar cold. Thick fur, wide weight-spreading paws and a fat layer all help in the cold, so the large ears are LEAST helpful.